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"On my honor, I have neither given nor received any unauthorized and inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

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Question 3)

a) * Suppose, u is a unit vector in R^n.

* H is the $n \times n$ symmetric matrix such that, $H = I - 2uu^T$, as u is a unit vector, so $\|u\| = 1$

$$H^2 = H \cdot H = (I - 2uu^T)(I - 2uu^T)$$
$$= I^2 - I.2uu^T - I.2uu^T + 4uu^Tuu^T$$
$$= I - 2uu^T - 2uu^T + 4u.\|u\|^2u^T$$
$$(\|u\|^2 = u^Tu,\ \|u\| = +\sqrt{u^Tu},\ \|u\| = 1)$$

$$H^2 = I - 4uu^T + 4uu^T$$
$$H^2 = I$$

b) $$HH^T = (I - 2uu^T)(I - 2uu^T)^T$$
$$= (I - 2uu^T)(I^T - (2uu^T)^T) \quad * (A-B)^T = A^T - B^T$$
$$= (I - 2uu^T)(I - 2(u^T)^Tu^T) \quad * (AB)^T = B^TA^T$$
$$= (I - 2uu^T)(I - 2uu^T) \quad * (u^T)^T = u$$
$$= I - 4uu^T + 4u\|u\|^2u^T$$
$$= I - 4uu^T + 4uu^T$$
$$= I$$

$$H^TH = I$$

so, $HH^T = H^TH = I$

* H is an orthogonal matrix.

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c) $\boxed{Av = \lambda v}$

$$Hv = (I - 2ww^T)v$$
$$= Iv - 2ww^Tv$$
$$= v - 2v \cdot \|v\|^2$$

* I is identify matrix, so 1 is the only eigen value of I and, $Iv = v$, and, $\|v\| = +\sqrt{v^Tv}$ v is a unit vector, so $\|v\| = 1 = +\sqrt{v^Tv}$ and $v^Tv = \|v\|^2 = 1$.

$$Hv = v - 2v \cdot 1 = -v = (-1)v$$

so $Hv = \lambda v$, where $\boxed{\lambda = -1}$

v is an eigen vector of H and corresponding eigen value of v is $\lambda = -1$.

d) given $H = (I - 2ww^T)$

$$H^T = (I - 2ww^T)^T$$
$$= I^T - 2(ww^T)^T$$
$$= I - 2(w^T)^T w^T \quad \text{so, H is a symmetric matrix.}$$
$$= I - 2ww^T = H$$

theorem,

$H = Q \Lambda Q^T$ (Q is an orthogonal matrix)

$$= Q \Lambda Q^{-1}$$
$$= (Q^{-1})^{-1} \Lambda Q^{-1}$$
$$= P^{-1} \Lambda P$$

so, H is diagonalizable matrix.

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